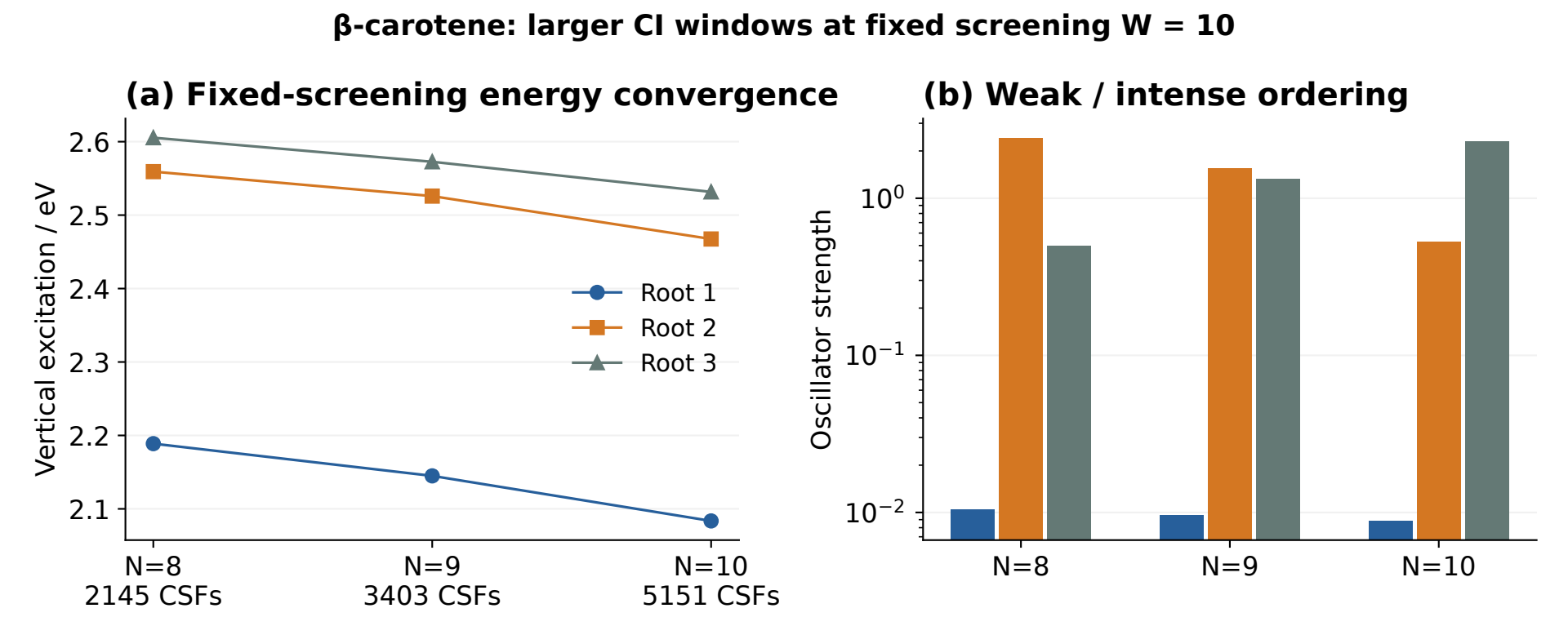
β-carotene: larger CI windows at fixed screening W = 10
(a) Fixed-screening energy convergence
Vertical excitation / eV
2.6
2.5
2.4
2.3
2.2
2.1
Root 1
Root 2
Root 3
N=8
2145 CSFs
N=9
3403 CSFs
N=10
5151 CSFs
(b) Weak / intense ordering
Oscillator strength
10^{0}
10^{-1}
10^{-2}
N=8
N=9
N=10